

LIRUI WANG

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EDUCATION

Massachusetts Institute of Technology

Sep 2021 -

University of Washington, Seattle

Sep 2016 - Mar 2021

Bachelor of Science, Electrical Engineering (Honors), College of Engineering

Bachelor of Science, Computer Science (Honors), College of Arts and Sciences

Bachelor of Science, Mathematics (Minor), College of Arts and Sciences

Master of Science, Computer Science, College of Arts and Sciences

summa cum laude

Advisor: Professor Dieter Fox

RESEARCH PROJECTS

6D Instance Grasping with Point Clouds

- Investigated actor-critics methods with planner demonstrations and goal auxiliary tasks, to learn closed-loop control policies for 6D robotic grasping using point clouds.
- Achieved over 90% success rates on grasping various unseen objects in both simulation and real world.

Joint Motion and Grasp Planning

- Proposed an optimization-based integrated planner that optimizes manipulation trajectory with on-line grasp synthesis and selection.
- Demonstrated robust and smooth motion plans for grasping in cluttered scenes in the real world and outperformed several fixed-endpoint baselines.

Pose Tracking with the Deep Iterative Matching Network

- Evaluated DeepIM on YCB Video dataset and improved training on synthetic data by generating diverse pose distribution and realistic occlusions.
- Extended DeepIM to achieve state-of-the-art pose tracking accuracy and AR applications.

WORK EXPERIENCE

Seattle, WA

Research Assistant, UW Robotics and State Estimation Lab

- Worked on robot manipulation research under the supervision of Dr. Yu Xiang and Prof. Dieter Fox at UW and NVIDIA.

Seattle, WA

Applied Research Intern, NVIDIA

- Investigated a traffic predictive model to simulate reactive, multi-modal traffic and integrated into the NVIDIA simulation framework for model-based reinforcement learning and planner validation.

Seattle, WA

Teaching Assistant, UW CSE Department

- Assisted teaching for computer vision (CSE 455), deep learning (CSE 543), and artificial intelligence (CSE 473) at UW CSE department.

Shenzhen, China

Research Engineer Intern, DJI Technology

- Implemented drone functions such as take-off, tag detection, and multi-robot communication and navigation systems such as SLAM and high-level state machines.

MISCELLANEOUS PROJECTS

- **ARUW Robomaster Competition:** Managed a team of 10 students to design computer vision algorithms for an autonomous robot battle, including robots with onboard NVIDIA TX1 and drones with DJI Manifold with Guidance, in the RoboMasters competition hosted by DJI.
- **Baxter Hand-Eye Coordination:** Developed software including camera models, hand-eye calibrations, robot kinematics and dynamics, shader-based rendering for Baxter to manipulate objects.
- **Microsoft Server Mover Project:** Built an autonomous mobile manipulator, a UR5 arm mounted on a server forklift, to pull out servers from a server rack.
- **Survey and Empirical Evaluation of Unsupervised Flow and Depth Methods:** Studied structure from motion methods that jointly learn 3D vision tasks such as depth and flow.
- **Survey and Empirical Evaluation of Attacks and Defense for Robust Classification:** Investigated example meta-reweighting and adversarial training techniques to robustify a fossil classification task and surveyed on attacks and defense of adversarial examples for neural networks.
- **Survey on Continuous Trajectory Optimization:** Studied formulation of trajectory optimization as functional gradient descent using Reproducing Kernel Hilbert Space and as probabilistic inference on a factor graph using Gaussian Process.
- **Survey on Representation Learning for RL:** Study works that study theoretical conditions and guarantees of representation learning for RL sample complexity and robotic applications.

HONORS

- MIT EECS Fellowship (Xianhong Wu Fellow), MIT, *2021 - 2022*
- Herbold Data Science Fellowship, UW, *2020 - 2021*
- Patricia Lynch & Eugene Russell Memorial Computer Science Scholarship, UW, *2018 - 2019*
- Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship, UW, *2017 - 2018*
- Tau Beta Pi Scholarship, UW, *2017 - 2019*
- Annual Dean's List, UW, *2016 - 2020*

PUBLICATIONS

- **Goal-Auxiliary Actor-Critic for 6D Robotic Grasping with Point Clouds**
Lirui Wang, Yu Xiang, Dieter Fox
Under Review
- **Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection**
Lirui Wang, Yu Xiang, Dieter Fox
In Proceedings of Robotics: Science and Systems (RSS), 2020

SELECTED COURSES

- **UW:** Deep Learning, Computer Vision, Computer Graphics, Modern Algorithm Toolbox, Robustness in Machine Learning, Interactive Learning, Probabilistic Robotics, Reinforcement Learning, Data Science, Theory of Deep Learning, Control Theory, Optimization and Learning for Control, Convex Optimization and Modeling, Numerical Analysis, Linear Analysis, Convex Algorithm and Analysis, High Dimensional Statistics, Statistical Learning, Signal Processing

SKILLS

- **Platforms:** Arduino, NVIDIA TX Series, STM32, Franka Panda, Baxter, UR5e, DJI Manifold
- **Languages:** Python, C++/C, Matlab, Java
- **Tools:** Git, CMake, OpenCV, Tensorflow, Pytorch, Latex, OpenGL, ROS, Ubuntu
- **Hardware:** 3D Printing, Laser Cutting, Machining, Soldering